Oswin Alex

+91 9324498843 | oswinalex1@gmail.com | oswin-alex | Alexoswin | oswinalex.site

EXPERIENCE

Software Engineer Sept. 2025 – Present

MIDGENIE AI LABS PRIVATE LIMITED — mple.ai Mumbai, India

- Develop and scale backend services for **mple.ai**, an **AI-driven sales training and agentic learning platform** serving **500K+ users** across realistic customer service and sales roleplay simulations.
- Cut **REST API** and **MongoDB aggregation** response times from **5–7 seconds to under 300 ms (95% faster)** by rewriting **20+ aggregation pipelines**, adding compound indexes, and root-causing production incidents end to end.
- Integrated **LLM agents, Retrieval-Augmented Generation (RAG)**, and a **host/worker multi-agent architecture** that sustains **10K+ conversational roleplay sessions per month** at sub-2 second first-response latency.
- Engineered a **rate-limited asynchronous messaging service** on **AWS SQS and Redis**, processing **1K+ messages/minute** with queue-based concurrency caps and caching that eliminated **100% of downstream overload failures**.
- Delivered and maintain seeedemaseekhelp.com, an AI-based edema score assessment platform.

PROJECTS

ReachConvert – AI-Powered Outreach Platform Link

- Built an open-source **AI-powered outreach platform** combining personalized email campaigns, AI voice calling, lead management, and campaign analytics in a unified workspace.
- Engineered automated **AI voice agents** for outbound calling and conversational lead engagement, alongside bulk email campaigns with personalized messaging and delivery tracking.
- Designed the platform with a scalable full-stack architecture supporting **AI agents, campaign automation, contact management, and real-time outreach tracking**.
- Tech Stack: **Next.js**, **NestJS**, **MongoDB**, **Redis**, **AWS SES**, **Twilio**, **Google Gemini**, **Tailwind CSS**

AI-Based Edema Score Assessment System Link | *mple.ai*

- Built a public **patient self-screening web platform** that scores **edema risk** from a **structured questionnaire** and returns an instant preliminary score with guidance on when to seek care.
- Integrated a **conversational AI doctor avatar** to guide patients through symptom questions in real time.
- Automated **assessment report generation** from questionnaire responses, giving patients a formatted summary of their score.
- Designed **REST APIs** for edema scoring, patient response processing, and report delivery.
- Tech Stack: **Next.js**, **NestJS**, **MongoDB**, **Tailwind CSS**

OPEN SOURCE CONTRIBUTIONS

Frappe Framework – OAuth2 Refresh Token Fix PR #36145

- Resolved a defect in the **Frappe Framework** (**8K+ star** open-source repository) where **OAuth2 refresh token requests returned 403 Forbidden**, breaking token renewal for every OAuth2 client integration.
- Diagnosed the root cause across **2 token validation methods** that loaded **OAuth Bearer Token** documents without **ignore_permissions=True**, failing under guest context.
- Patched validation by passing **ignore_permissions=True** and setting **request.user = otoken.user**, restoring **100% of refresh token requests** to success and full **OAuth2 RFC 6749** compliance.
- Authored automated test cases for the refresh token grant and verified **0 regressions** across existing OAuth2 authentication flows.

EDUCATION

Don Bosco Institute of Technology Dec. 2021 – May 2025
Bachelor of Engineering in Information Technology (CGPA- 7.25) Mumbai

ACHIEVEMENTS

Winner – Smart India Hackathon 2023 (SIH1366) Dec. 2023
Problem Statement: Centralized System for De-addiction Centers Udaipur

- Led a **6-member team** to **1st place** among **10,000+ competing teams**, shipping a centralized full-stack system with real-time patient tracking and **RBAC across 4 user roles** in **36 hours**.

CERTIFICATIONS

AWS Academy Cloud Foundations Link Apr. 2025
Issued by Amazon Web Services

- Completed a foundational cloud program spanning **10+ core AWS services** (EC2, S3, IAM, VPC, RDS, Lambda) plus cloud security, the shared responsibility model, pricing, and compliance frameworks.

SKILLS

Languages: JavaScript, TypeScript, Python

Backend: Node.js, NestJS, Express.js, REST APIs, WebSockets

Frontend: Next.js, React.js, React Native, React Native WebView

Database: MongoDB, Redis, PostgreSQL

AI/ML: Generative AI, LLM Integration, Multi-Agent Systems, RAG, Retrieval-Augmented Generation

Cloud: AWS S3, AWS SES, AWS SQS, AWS CloudFront, Google Cloud Run, Google Compute Engine (GCE), Google Cloud Scheduler

Tools: Docker, Git, GitHub, CI/CD, Linux